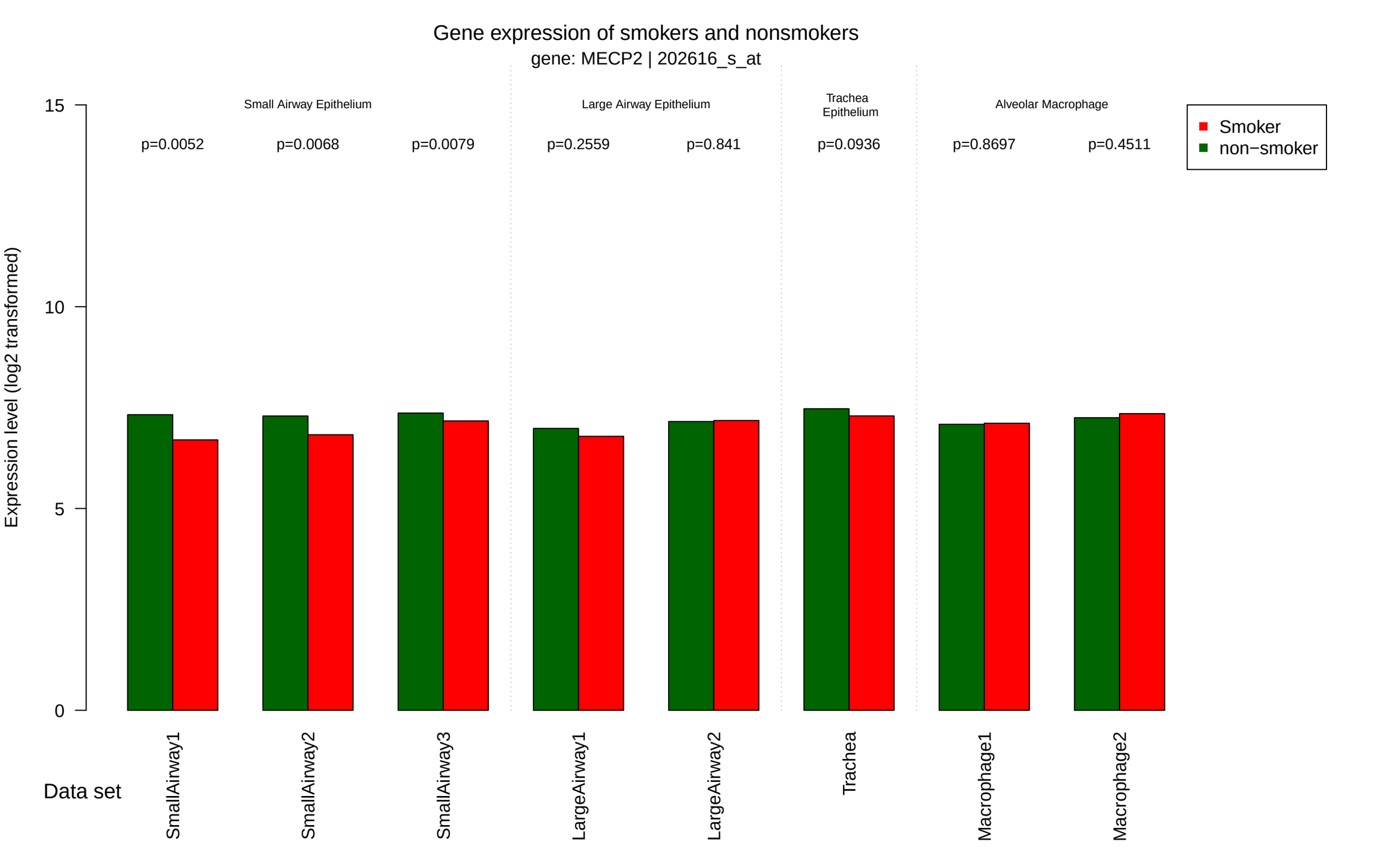
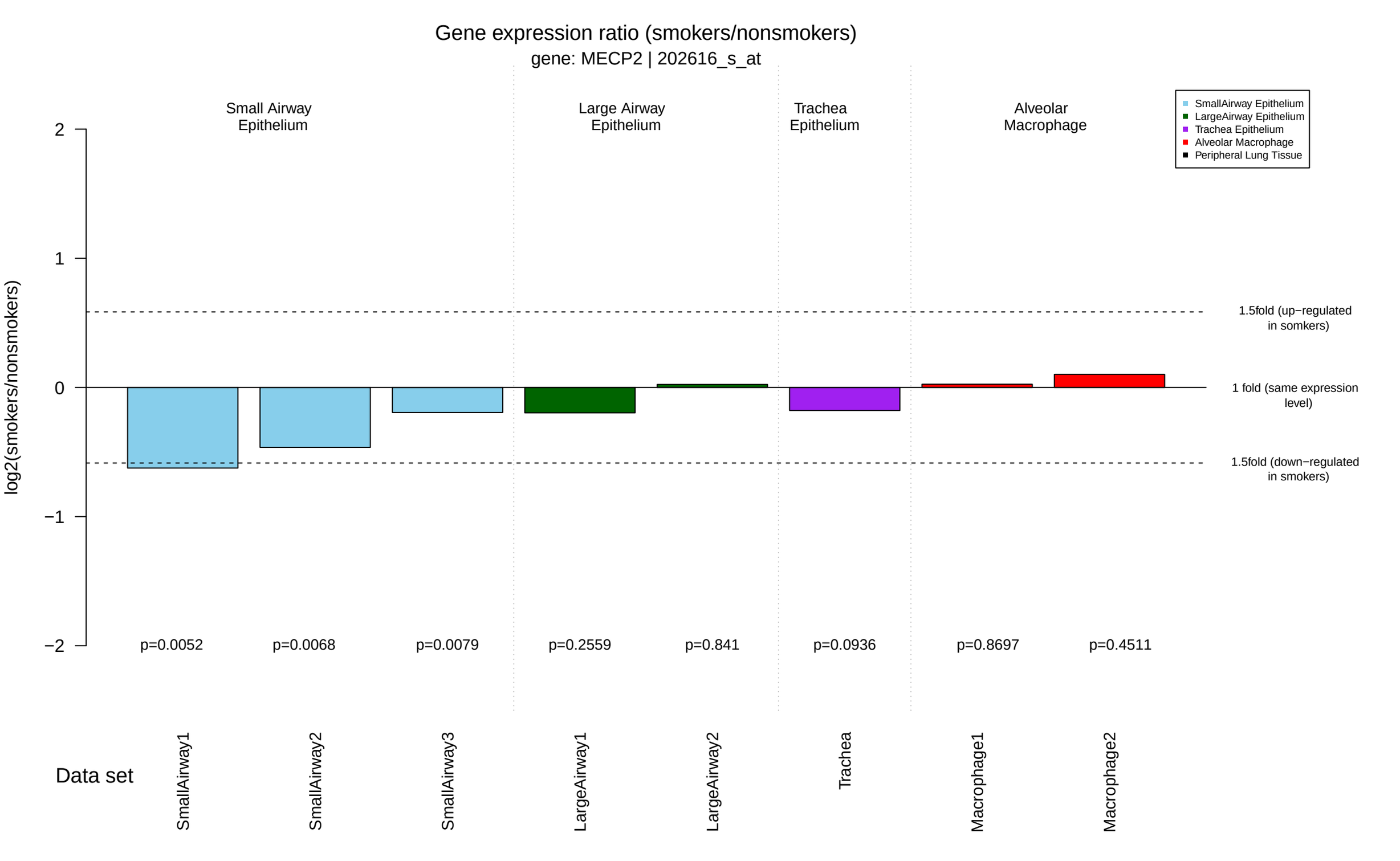
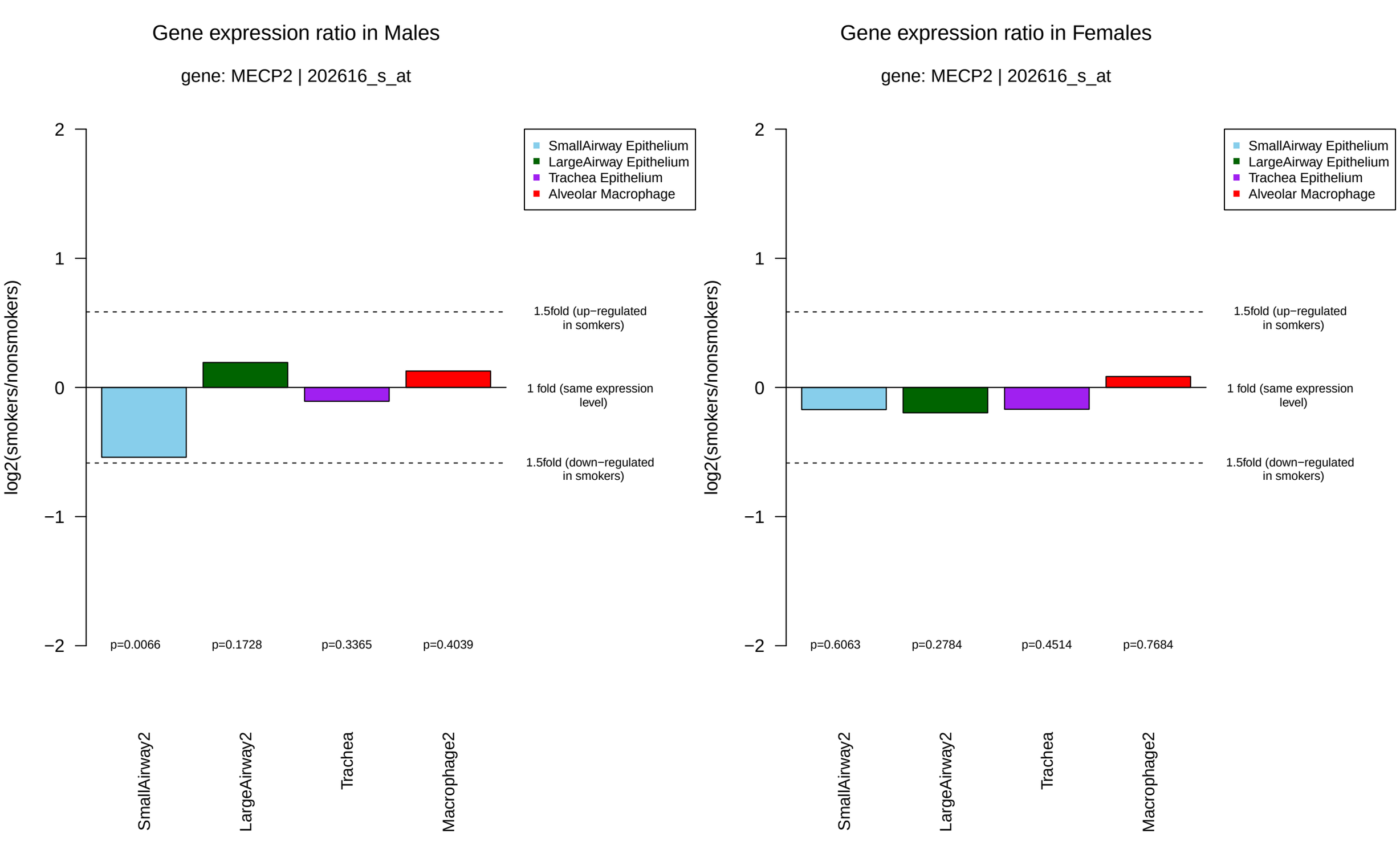
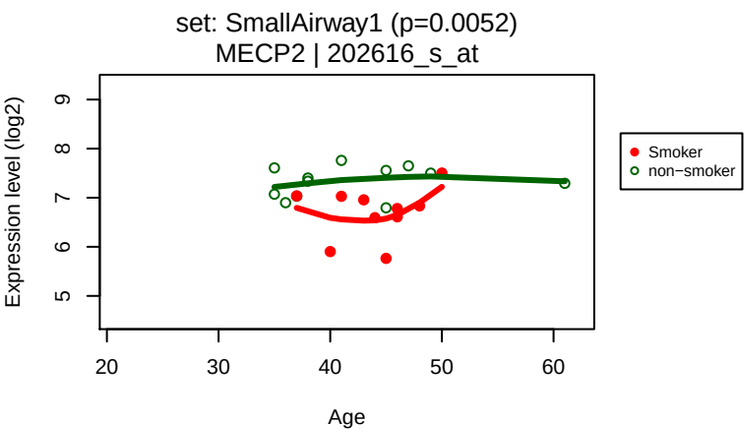
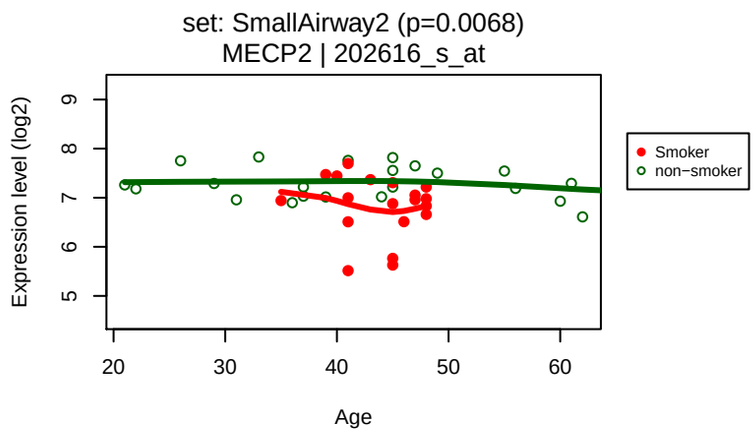
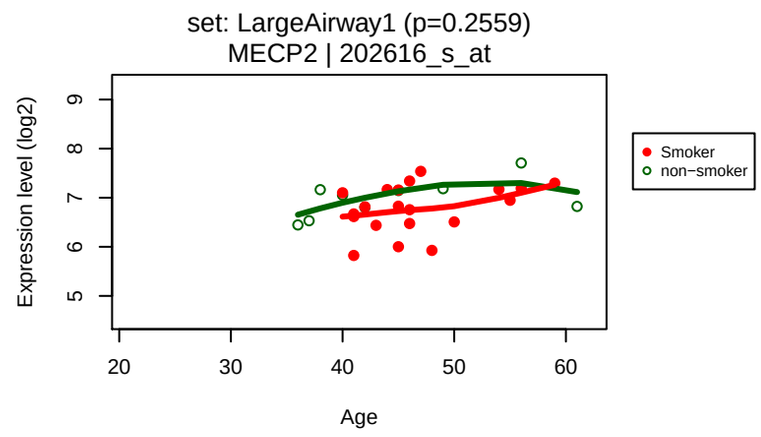

Gene expression of smokers and nonsmokers
gene: MECP2 | 202616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0052
p=0.0068
p=0.0079
p=0.2559
p=0.841
p=0.0936
p=0.8697
p=0.4511
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

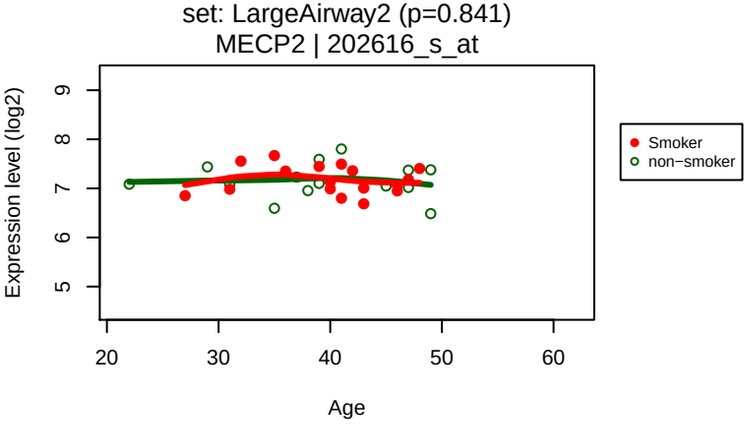

Gene expression ratio (smokers/nonsmokers)
gene: MECP2 | 202616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0.0068
p=0.0079
p=0.2559
p=0.841
p=0.0936
p=0.8697
p=0.4511
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

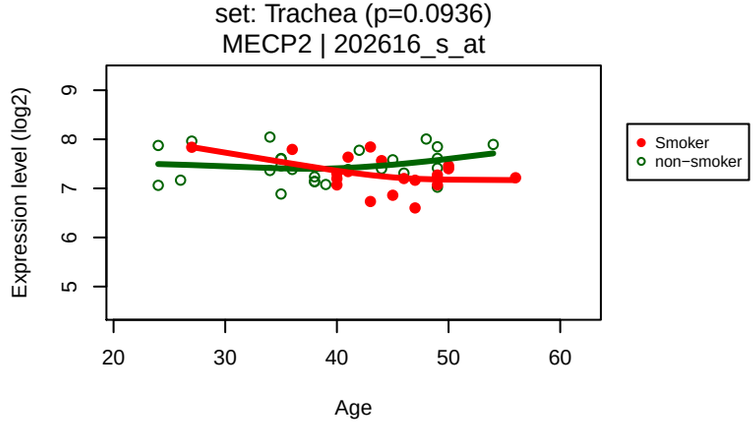

Gene expression ratio in Males
gene: MECP2 | 202616_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0066
p=0.1728
p=0.3365
p=0.4039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MECP2 | 202616_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6063
p=0.2784
p=0.4514
p=0.7684
SmallAirway2
LargeAirway2
Trachea
Macrophage2

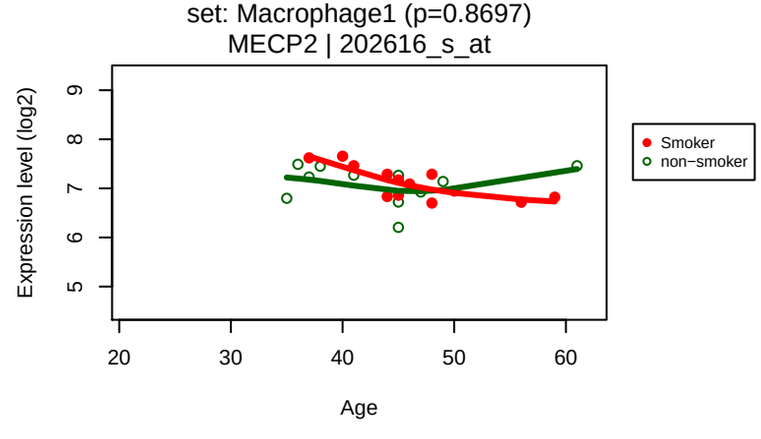

set: SmallAirway1 (p=0.0052)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker

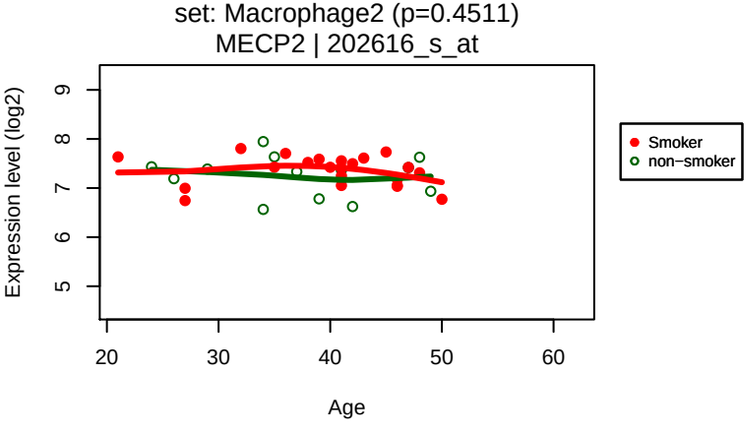

set: SmallAirway2 (p=0.0068)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker

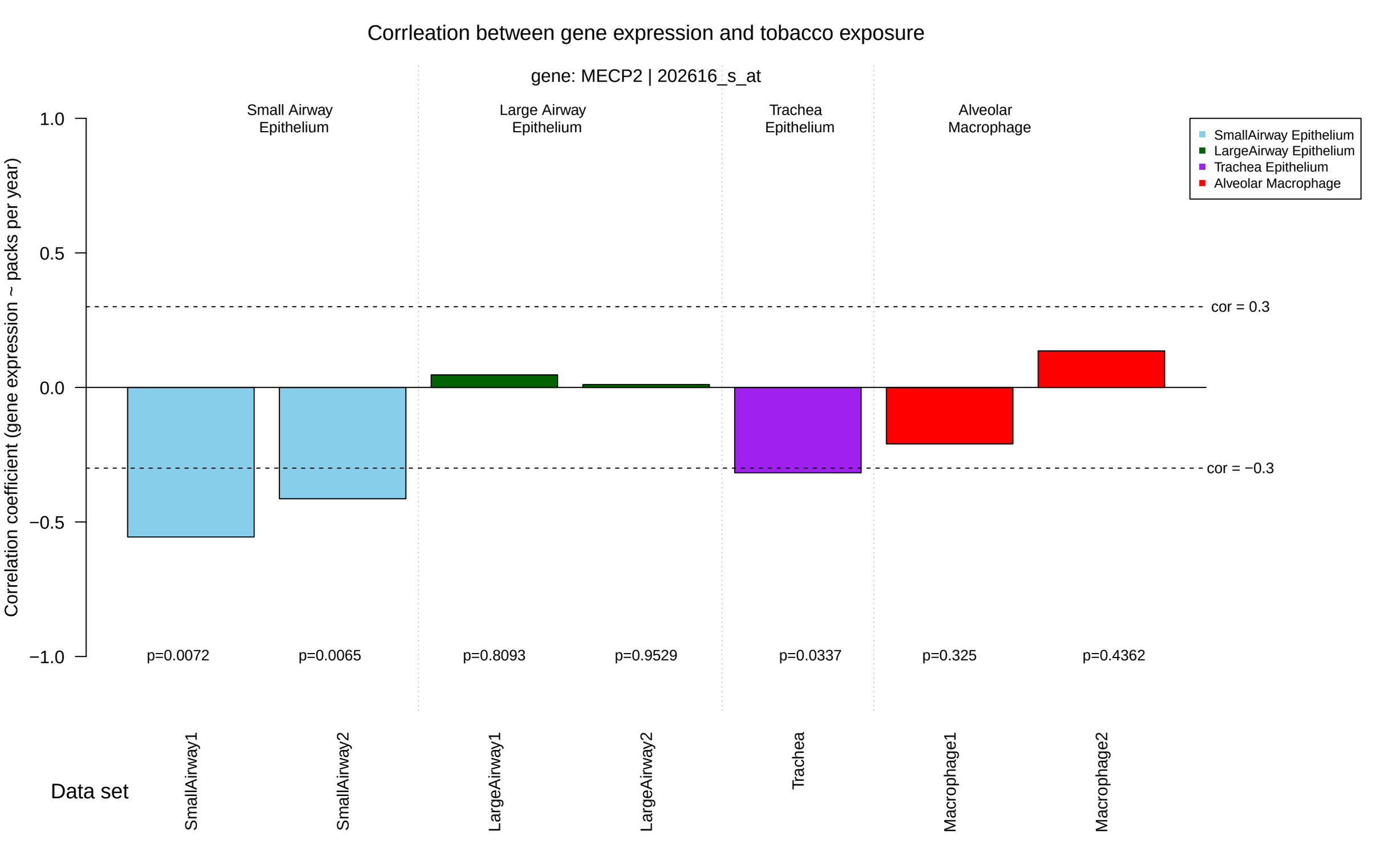

set: LargeAirway1 (p=0.2559)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.841)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0936)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8697)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4511)
MECP2 | 202616_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MECP2 | 202616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0072
p=0.0065
p=0.8093
p=0.9529
p=0.0337
p=0.325
p=0.4362
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2